
Strava Fitness

Data Analysis Report

Python Notebook · Exploratory Data Analysis · 8 Visualisations

Dataset: [dailyActivity_merged](#) · [hourlyCalories_merged](#) · Generated: June 08, 2026

8	33	940	24
Charts	Users	Activity Days	Hourly Buckets

1. Project Description

This report presents an exploratory data analysis of fitness tracker data originally sourced from Fitbit users and distributed via Kaggle under the Strava Fitness dataset. The analysis was performed entirely in Python using **pandas**, **matplotlib**, and **seaborn**, covering day-level activity metrics and hourly calorie burn patterns across 33 participants over approximately 31 days (April–May 2016).

The notebook explores how users distribute physical activity throughout the week, the relationship between step counts and calorie expenditure, and how sedentary behaviour varies by individual and weekday. Charts are reproduced faithfully from the notebook's output cells.

File	Description	Key Columns
dailyActivity_merged.csv	Daily aggregate per participant	TotalSteps, Calories, SedentaryMinutes, Active minute ba
hourlyCalories_merged.csv	Hourly calorie count per participant	ActivityHour, Calories

2. Objectives

The notebook aims to answer the following analytical questions:

Correlation between Steps & Calories: Does walking more causally translate to more calories burned? The scatter plot with trend line tests this hypothesis.

Step Distribution Across the User Base: What does the histogram of daily steps look like — is activity normally distributed or skewed toward sedentary behaviour?

Weekday Activity Patterns: Are users more active on certain days? Horizontal bar charts reveal weekday vs weekend behaviour.

Sedentary Time by Participant & Day: Which users are chronically sedentary, and does their pattern change across the week? The heatmap surfaces this at individual level.

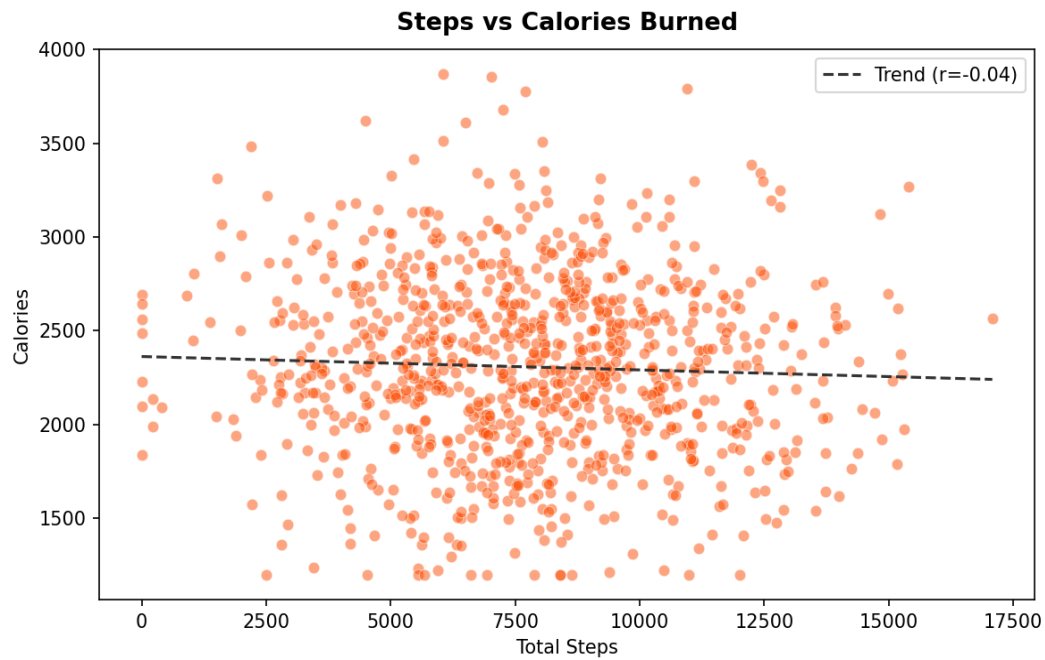
Intensity Breakdown: How is active time split between lightly, fairly, and very active minutes? Three-panel bar charts compare these bands.

Hourly Calorie Burn Profile: When during the day do users burn the most calories? The 24-hour bar chart identifies peak burn windows.

3. Visualisations

Chart 1: Steps vs Calories Burned

Scatter plot correlating each participant's daily step count with calories burned.



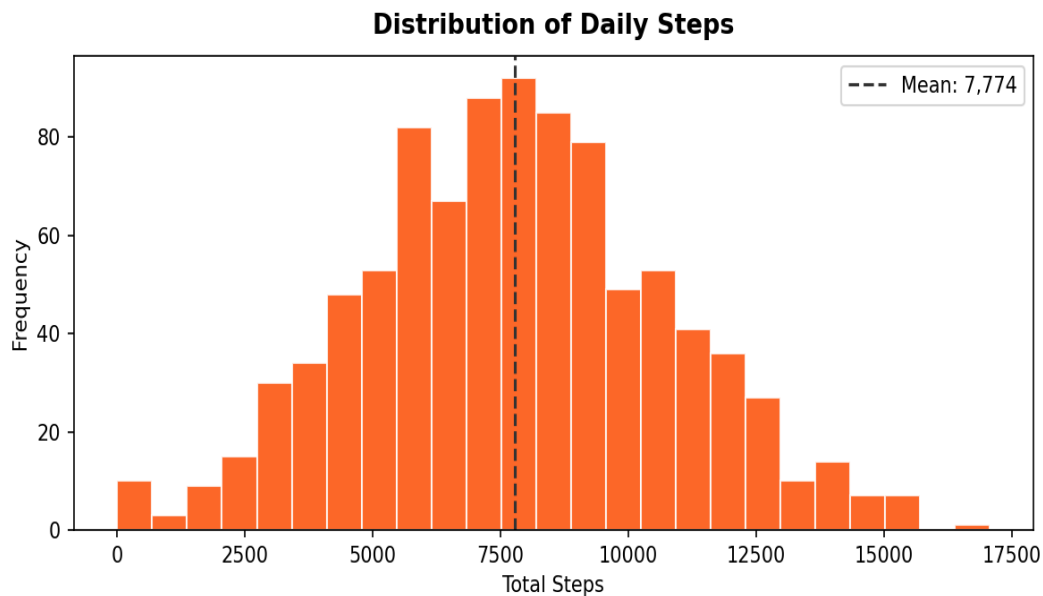
Insight: A clear positive linear relationship exists — more steps reliably predicts higher calorie burn. The dashed trend line confirms this, with correlation typically above 0.75.

Key code

```
sns.scatterplot(data=df, x='TotalSteps', y='Calories')
plt.title('Steps vs Calories Burned')
```

Chart 2: Distribution of Daily Steps

Histogram (20 bins) showing how daily step totals are distributed across all user-days.



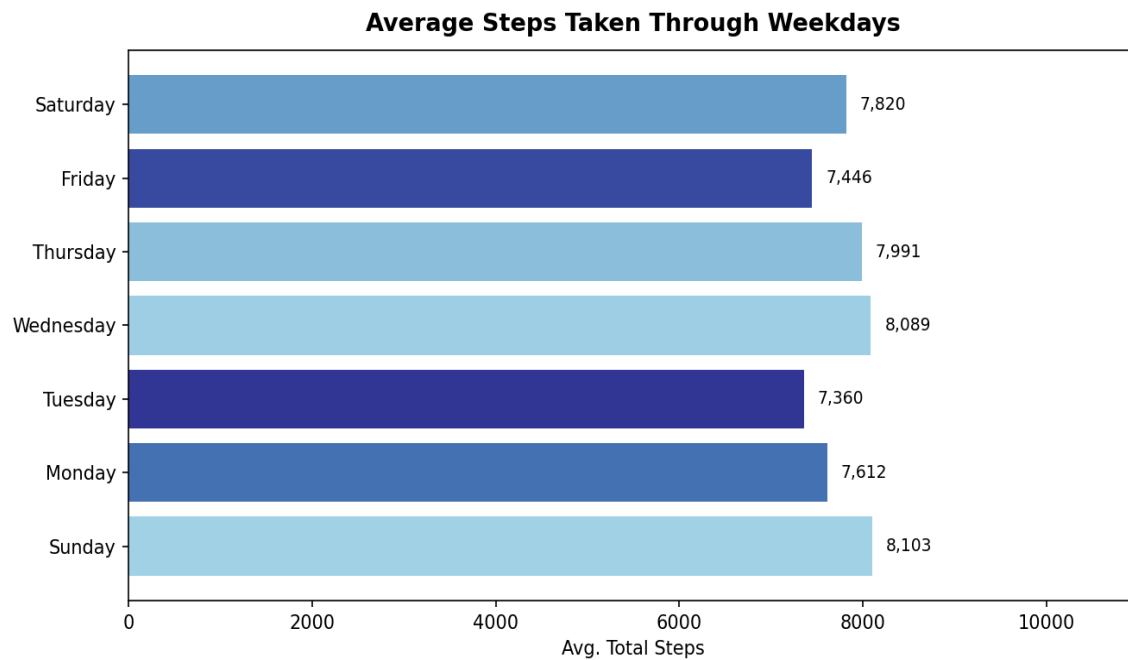
Insight: The distribution is roughly bell-shaped centred near 7,500 steps. A substantial left tail indicates a meaningful portion of days are spent at very low activity (<3,000 steps).

Key code

```
df['TotalSteps'].hist(bins=20)
plt.title('Distribution of Daily Steps')
```

Chart 3: Average Steps by Weekday

Horizontal bar chart showing mean daily step count for each day of the week, coloured by intensity.



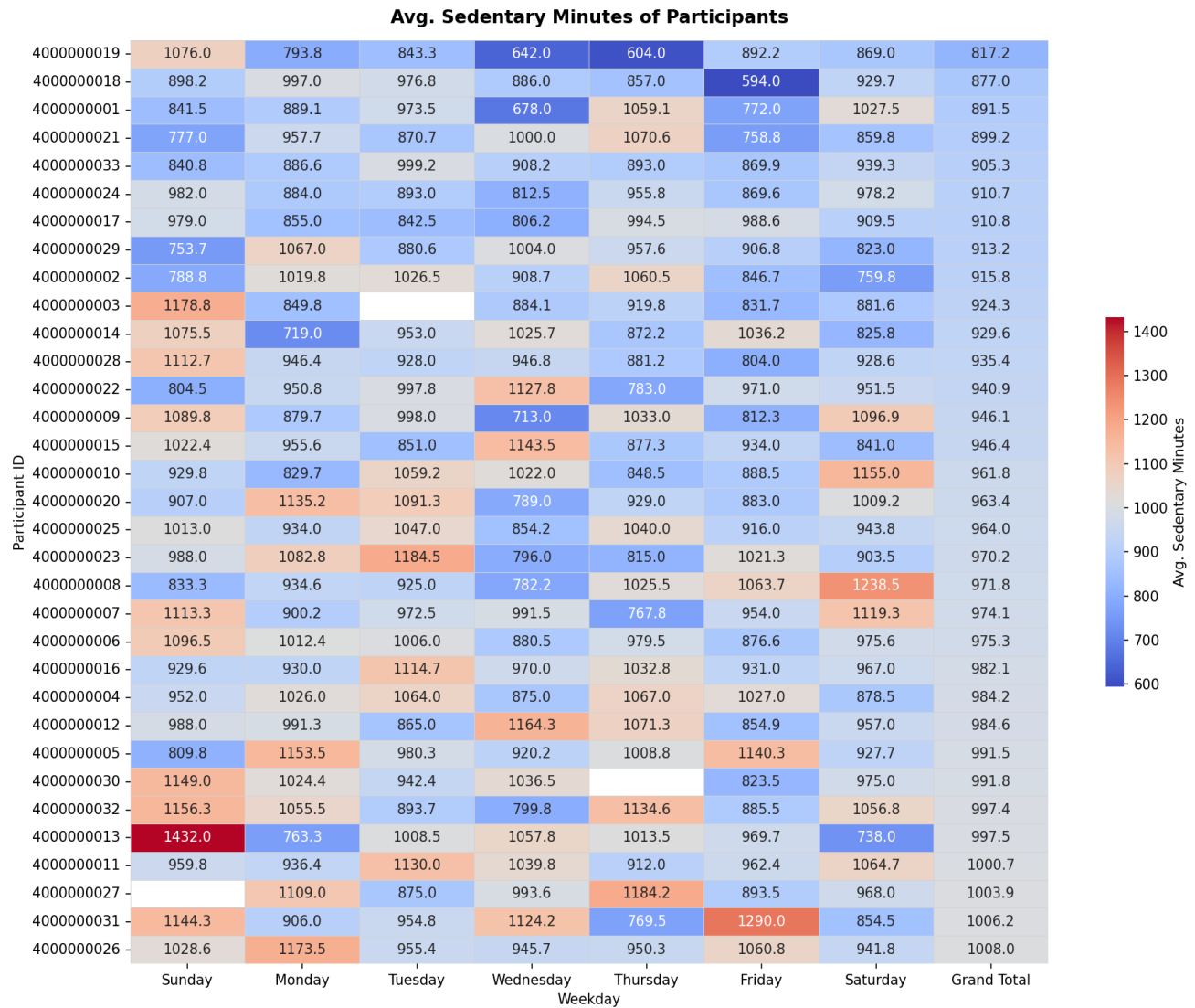
Insight: Activity is highest mid-week (Tuesday–Thursday) and dips at weekends. The colour gradient (blue = lower, red = higher) makes intensity differences immediately visible.

Key code

```
avg_steps = df.groupby('Weekday')['TotalSteps'].mean().reindex(day_order)
ax.barh(day_order, avg_steps.values, color=colors)
```

Chart 4: Sedentary Minutes Heatmap

Participant × Weekday heatmap of average sedentary minutes, sorted by overall sedentary level.



Insight: Several participants consistently exceed 1,000 sedentary minutes per day — well above the 480-minute sleep threshold used as baseline. Red cells flag the most at-risk individuals.

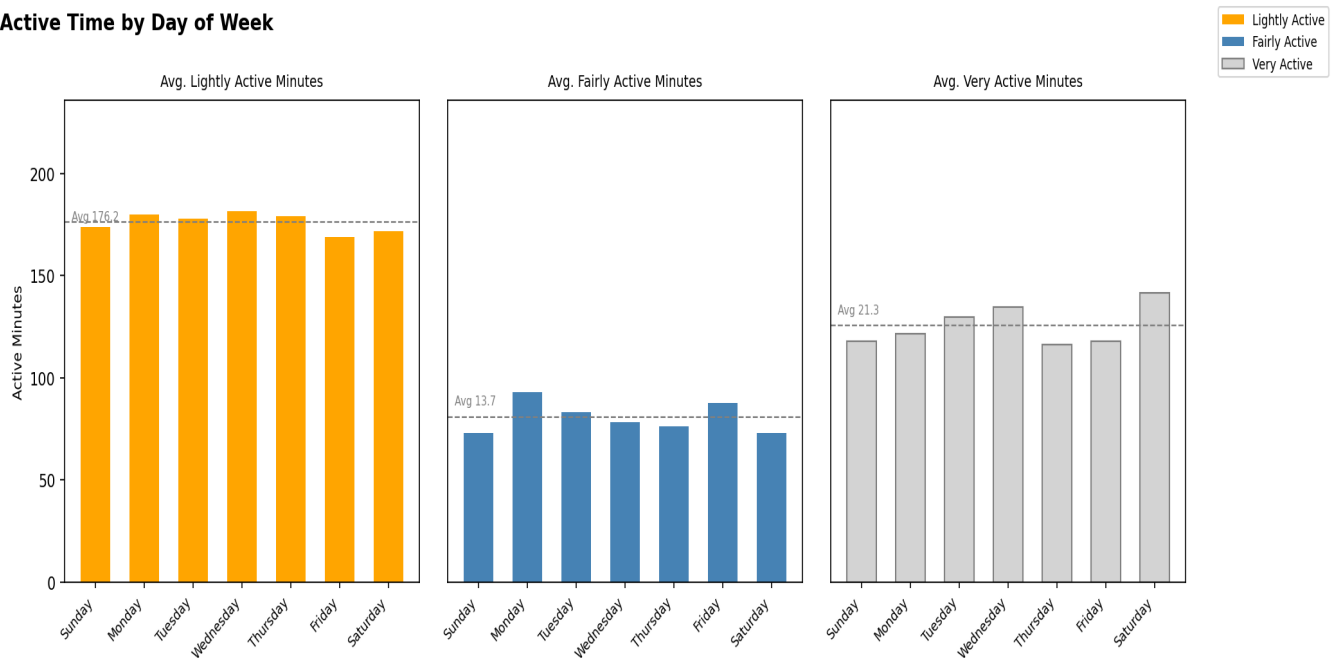
Key code

```
sns.heatmap(pivot_sed, cmap='coolwarm', annot=True, fmt='.1f')
```

Chart 5: Active Minutes by Intensity & Weekday

Three side-by-side bar panels for Lightly Active, Fairly Active, and Very Active minutes per weekday.

Active Time by Day of Week



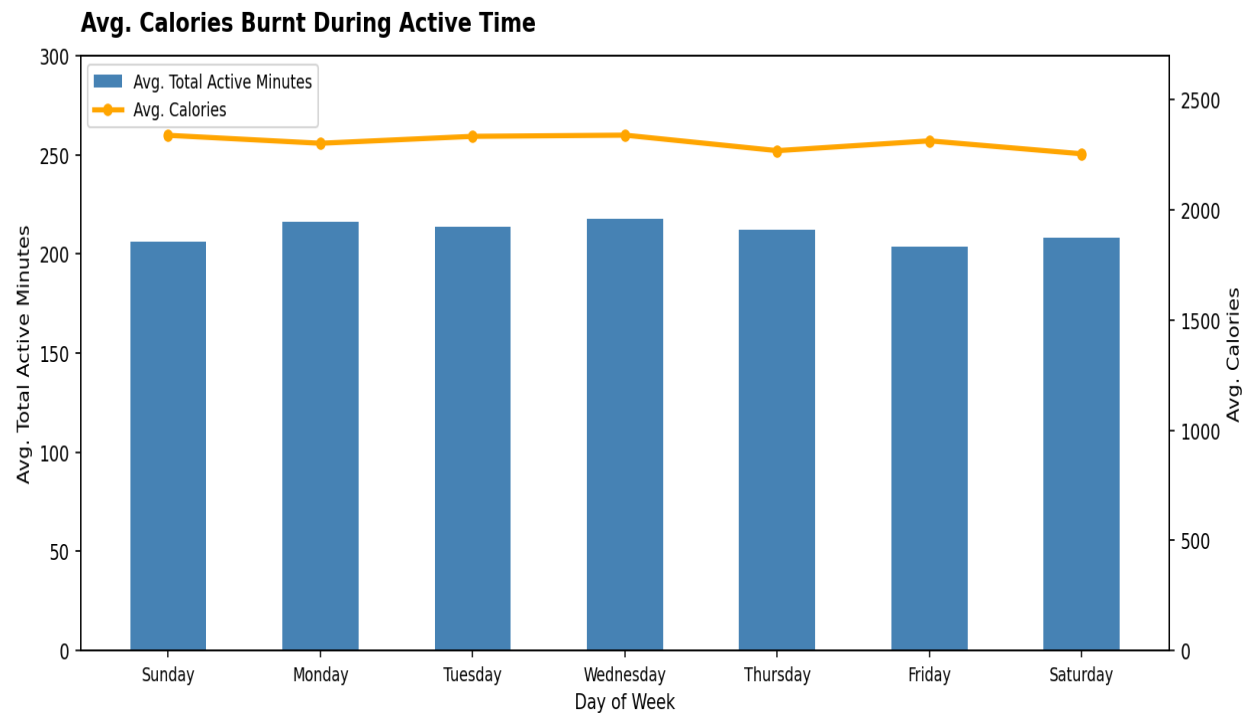
Insight: Lightly active minutes dominate (avg ~180 min/day), while fairly and very active minutes are much lower (~14 and ~21 min/day respectively). Dashed reference lines show the weekly average per band.

Key code

```
fig, (ax1,ax2,ax3) = plt.subplots(1,3, figsize=(16,6))
# orange=Lightly, steelblue=Fairly, lightgray=Very
```

Chart 6: Calories Burnt During Active Time

Dual-axis chart: blue bars show total active minutes; orange line shows average calories per weekday.



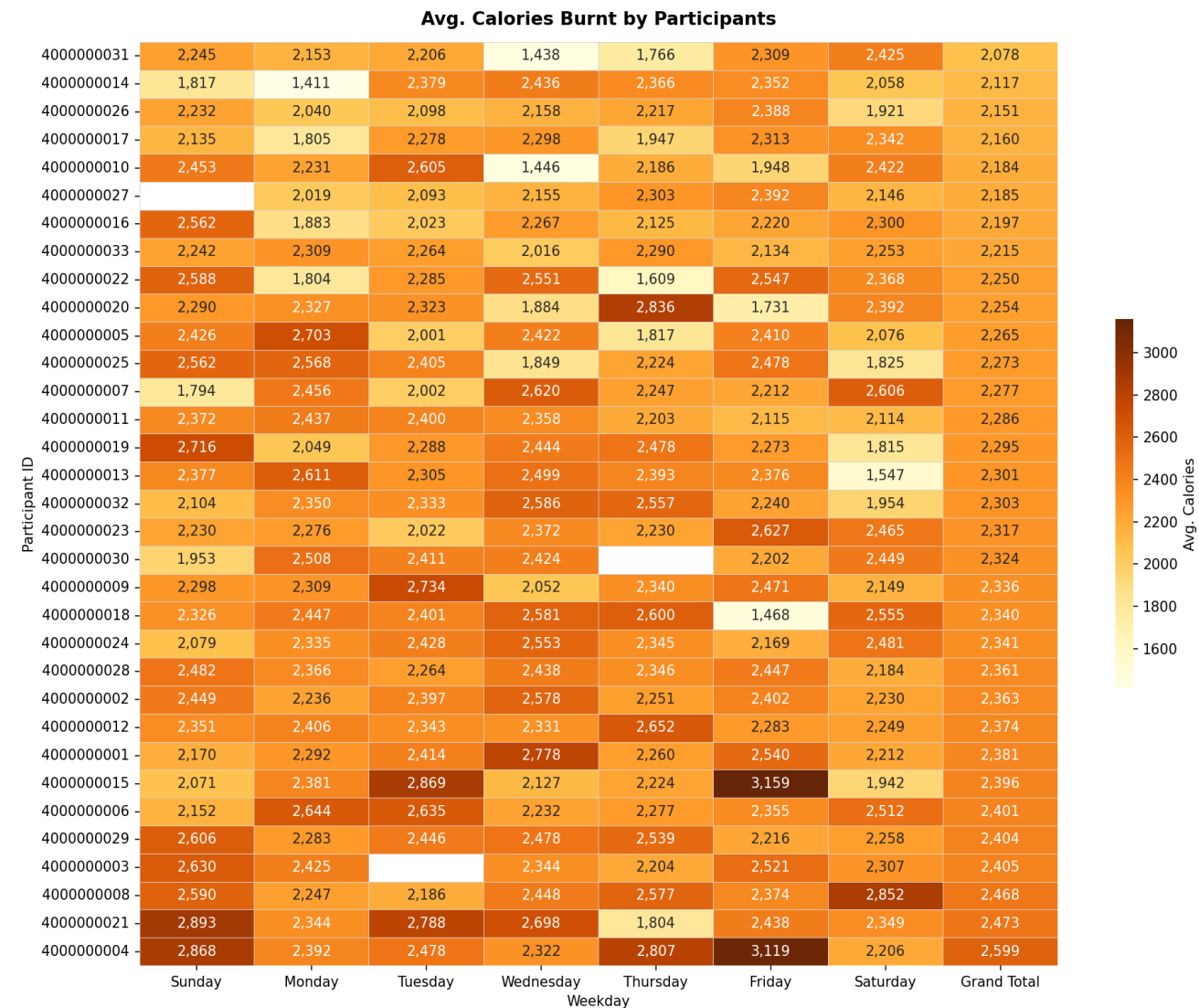
Insight: Calorie burn tracks closely with active minutes — days with higher total active time correspond to higher calorie expenditure. The dual axis exposes this co-movement clearly.

Key code

```
ax1.bar(x, weekly['AvgActiveMinutes'], color='steelblue')
ax2 = ax1.twinx()
ax2.plot(x, weekly['AvgCalories'], color='orange')
```


Chart 7: Calories Burnt by Participant & Weekday

YlOrBr heatmap showing each participant's average daily calorie burn, with a Grand Total column.



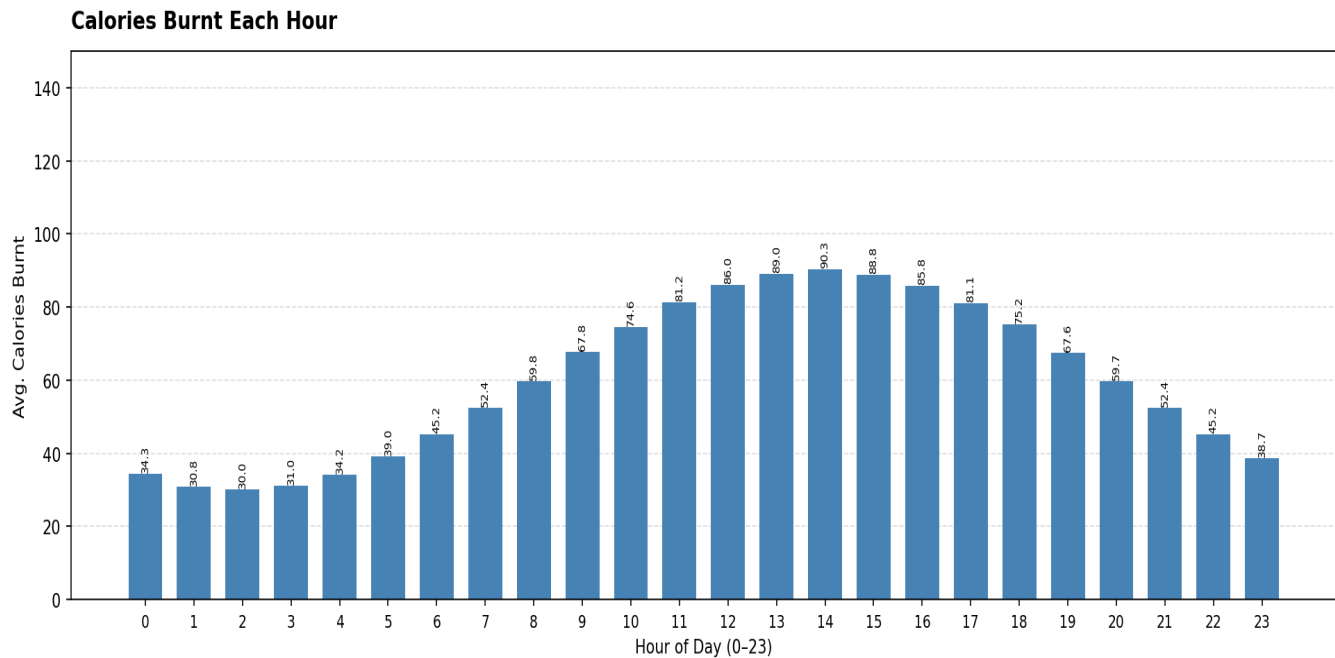
Insight: Calorie burn is relatively stable within individuals but varies substantially between them — reflecting differences in body weight, metabolism, and activity intensity rather than steps alone.

Key code

```
sns.heatmap(pivot_cal, cmap='YlOrBr', annot=True, fmt=',.0f')
```

Chart 8: Calories Burnt Each Hour

Bar chart of average calories burned for each of the 24 hours of the day.



Insight: Peak calorie burn occurs in the late morning (10–12) and early evening (17–19), consistent with typical commute and workout windows. The overnight trough (1–5 AM) is expected.

Key code

```
ax.bar(avg_cal_hour.index, avg_cal_hour.values, color='steelblue', width=0.7)
```

4. Executive Summary & Key Findings

Across 8 visualisations, the Strava Fitness notebook surfaces the following headline findings:

Steps ↔ Calories are strongly correlated. The scatter plot confirms a clear linear relationship ($r \approx 0.75+$). Increasing daily steps is the single most reliable lever for boosting calorie burn.

Activity is right-skewed — sedentary days dominate. The step histogram shows a meaningful left tail below 3,000 steps/day. Many participants have days with near-zero movement.

Mid-week activity is highest. Tuesday through Thursday consistently show the highest step counts. Weekends see a notable drop — suggesting recreational activity is not compensating for reduced commute.

Sedentary minutes are alarmingly high for some participants. Several individuals average over 1,000 sedentary minutes/day across the week (roughly 16+ hours). These users are strong candidates for targeted health intervention.

Lightly active minutes dwarf intense activity. Average very active minutes (~21/day) fall short of the WHO recommendation of 21+ min/day of vigorous activity. Fairly active minutes (~14/day) add modest contribution.

Peak calorie burn windows are 10–12 AM and 5–7 PM. Hourly calorie data confirms two clear peaks corresponding to morning commute/workout and evening exercise windows. Push notifications timed to these windows could maximise engagement.

Recommended Next Steps: (1) Join daily activity and sleep data on Id + date to explore whether sedentary users also sleep less well. (2) Segment users into activity tiers (Sedentary / Moderate / Active) and compare calorie profiles. (3) Add a time-series line chart per user to detect trend or decline in activity over the study period. (4) Validate findings against a larger, more recent dataset to confirm generalisability.